NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer and Information Systems Engineering

SOFTWARE REQUIREMENTS SPECIFICATION

Computer Vision and AI-Based Construction Site Monitoring via Satellite Imagery

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1. Introduction

1.1 Document Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Computer Vision and AI-Based Construction Site Monitoring via Satellite Imagery system. It defines the expected system behaviour, interfaces, data requirements, constraints, and interaction models to guide design, implementation, and verification.

1.2 Product Scope

The **Computer Vision and AI-Based Construction Site Monitoring via Satellite Imagery** system aims to monitor construction progress by detecting physical changes in construction sites using satellite imagery, computer vision, and artificial intelligence. The system will allow authorized users, including builders, owners, and site supervisors, to register construction sites by providing the site name, geographical coordinates, and construction start date. Using Google Earth Engine, the system will retrieve available Sentinel-2 satellite images for the selected location and date range, preprocess them, and compare images captured at different times to identify and highlight changed areas.

The system will display detected changes, estimated changed areas, and AI-generated insights through a user-friendly dashboard. It will display the current detected changes for a selected monitoring check and allow users to generate and download reports. The project focuses on **site-level change detection and progress monitoring** and will not determine exact construction completion percentages, assess structural safety, or replace physical site inspections. Results will depend on image resolution, availability, cloud cover, and model accuracy.

1.3 Intended Audience and Document Overview

This document is intended for the project team, software and AI developers, reviewers, instructors, and other stakeholders who need to understand the system requirements. Chapter 1 introduces the project; Chapter 2 describes the product context and stakeholders; Chapter 3 specifies interfaces, user and system functional requirements, and behavioural models; Chapter 4 defines non-functional requirements; Chapter 5 records any additional requirements; and the appendices describe the data dictionary and traceability information.

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
AI	Artificial Intelligence
CV	Computer Vision
DFD	Data Flow Diagram

Term	Definition
FR	Functional Requirement
GEE	Google Earth Engine
IoU	Intersection over Union
NFR	Non-Functional Requirement
SFR	System Functional Requirement
SRS	Software Requirements Specification
Sentinel-2	Earth-observation satellite mission providing multispectral imagery
UFR	User Functional Requirement

1.5 Document Conventions

Requirements are identified by stable IDs where available. "Shall" indicates a mandatory requirement. Tables are used to describe detailed system requirements, including function, inputs, outputs, dependencies, actions, preconditions, postconditions, and side effects. Requirement IDs from the supplied report are preserved in this draft; inconsistencies and gaps should be reviewed before final submission.

1.6 References and Acknowledgments

1. Python 3.11+ Documentation
2. Django Documentation
3. Supabase Documentation
4. Google Earth Engine Documentation
5. Copernicus Sentinel-2 Satellite Imagery Documentation
6. StarUML — UML Modeling Tool used to create Use Case Diagrams, Sequence Diagrams, and Data Flow Diagrams (DFDs).
7. ChatGPT — AI tool used for assistance in SRS structuring, documentation, and requirement refinement.
8. Google Gemini — AI tool used for assistance in research, idea development, and documentation.
9. Class Diagram and Use Case Diagram — Reference material.
10. Software Requirements Specification (SRS) Template — Reference structure provided for the project.

2. Overall Description

2.1 Product Perspective

The system is an internet-based application for site-level construction change monitoring. Users register a site using a name, bounding-box coordinates, and construction start date. The system requests suitable Sentinel-2 imagery through Google Earth Engine, compares imagery from different dates, generates a change mask and estimated changed-area measurements, presents AI-generated observations, and supports monitoring report generation and download.

The system depends on external satellite imagery and geospatial services. Results depend on image availability, spatial resolution, cloud cover, image quality, and model performance. The system does not determine exact construction completion percentages, assess structural safety, or replace physical site inspections.

2.2 Product Functionality

- Account registration, login, and logout.
- Construction site registration, coordinate validation, map confirmation, and site details viewing.
- Site access requests and creator approval or rejection.
- Retrieval of suitable Sentinel-2 imagery through Google Earth Engine.
- Image preparation, change detection, generation of a change mask, and calculation of estimated changed area.
- Storage and retrieval of monitoring records and result references.
- Dashboard display of sites, monitoring status, imagery, detected changes, measurements, and AI observations.
- Generation and download of monitoring reports.
- Notifications for invalid input, unavailable imagery, processing failures, and external-service or database errors.

2.3 Users and Characteristics

Direct users are builders, construction site owners, and site supervisors. The supplied report states that registered users have the same general application permissions, while a site creator may approve or reject access requests for their own site.

Indirect stakeholders include the Software Engineer, AI Engineer, and System Administrator. Domain stakeholders include Google Earth Engine and the Sentinel-2 satellite imagery provider.

2.4 Operating Environment

The application is intended to be accessed through a modern web browser over the internet. The backend requires access to a database, Google Earth Engine services, and a processing environment capable of accessing satellite image pixels and running the selected computer-vision method. Exact deployment operating system, browser versions, memory, and storage minimums should be confirmed against the implementation.

2.5 Design and Implementation Constraints

- Satellite image availability varies by location, date range, cloud cover, and image quality.
- Sentinel-2 imagery has finite spatial resolution; results must not claim finer spatial detail than the source data supports.
- Google Earth Engine usage is subject to its access, quota, and service policies.
- The processing pipeline must access actual image pixel data; storing only image references is insufficient for model inference.
- Generated masks and other output files may be stored separately from database metadata, with references recorded in the database.
- The application requires internet connectivity for its web interface and external imagery services.

2.6 User Documentation

The final system should provide brief guidance for account creation, site registration, coordinate entry and confirmation, running an image comparison, interpreting change measurements and AI observations, viewing saved results, and generating/downloading reports. The actual user guide can be attached or linked when prepared.

2.7 Assumptions and Dependencies

- Users provide site coordinates and a construction start date; they do not upload satellite images.
- Google Earth Engine provides access to suitable Sentinel-2 imagery when available.
- The database stores site details, image references and metadata, monitoring records, measurements, and result references.
- Image pixels must be available to the computer-vision processing pipeline.

- All registered users have the same general application permissions; site creators have the site-specific ability to approve or reject access requests.
- AI-generated observations may be inaccurate and must remain grounded in the available image analysis and measurements.

2.8 Stakeholder Identification and Viewpoints

1. List of Stakeholders

- Construction Site Users: Builders, Owners, and Site Supervisors
- Software Engineer
- AI Engineer
- System Administrator
- Google Earth Engine
- Sentinel-2 Satellite Imagery Provider

2. Stakeholder Viewpoints

Viewpoint	**Stakeholder**	**Reason**
Direct	Construction Site Users (Builders, Owners, and Site Supervisors)	Register construction sites, monitor progress, view detected changes, review historical results, and generate reports. All users have the same application permissions.
Indirect	Software Engineer	Develop, maintain, and update the application and its backend services.
Indirect	AI Engineer	Develop, train, evaluate, and improve the computer vision model and AI-based analysis.
Indirect	System Administrator	Maintain system availability, manage technical configurations, and monitor system operation.
Domain	Google Earth Engine	Provide access to satellite imagery and geospatial processing capabilities, subject to service limitations and usage policies.
Domain	Sentinel-2 Satellite Imagery Provider	Supply satellite imagery used for construction-site monitoring, subject to image availability, spatial resolution, and data-usage conditions.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The application interface should provide registration and login screens, a dashboard, construction site registration with a map-based coordinate confirmation step, site details and access-request screens, image comparison controls, saved monitoring results, AI insights, and report generation/download controls.

3.1.2 Hardware Interfaces

The user-side hardware requirements consist of a device capable of running a modern web browser with a stable internet connection. The server-side hardware requirements depend on the selected computer-vision model and deployment environment and will be finalized during implementation.

3.1.3 Software Interfaces

The system interfaces with Google Earth Engine for Sentinel-2 image retrieval and geospatial processing; a database for user, site, monitoring, and report metadata; a processing environment for image preparation and change detection; and any configured AI component used to generate observations. Exact database, map provider, AI service, and storage implementation details should be recorded once finalized.

3.1.4 Communications Interfaces

The web application communicates with backend services over the internet using standard web protocols, normally HTTPS for production traffic. Backend services communicate with Google Earth Engine and the configured database using their supported APIs or drivers. Authentication credentials and sensitive data should be transmitted only through secure channels.

3.2 User Functional Requirements

3. User Functional Requirements

3.1. User Account Management

1. FR1: The system shall allow users to create an account by providing the required registration information.
2. FR2: The system shall allow registered users to log in using valid credentials.
3. FR3: The system shall allow authenticated users to log out of their accounts.

3.2. Data and Records Management

1. FR4: The system shall store and retrieve user information, construction site details, satellite imagery references, and monitoring results via database.
2. FR5: The system shall maintain records of monitoring operations and their associated results.
3. FR6: The system shall associate monitoring records with their corresponding construction sites.

3.3. Construction Change Detection

1. FR7: The system shall compare suitable satellite images captured at different times to identify changes within a registered construction site.
2. FR8: The system shall generate a change mask highlighting the areas where changes have been detected.
3. FR9: The system shall display the original satellite images alongside the corresponding change detection results.

3.4. AI-Based Construction Analysis

1. FR10: The system shall analyze detected changes to provide insights into construction site development over time.
2. FR11: The system shall summarize the detected changes in an understandable format for users.
3. FR12: The system shall present AI-generated observations based on the available image analysis results.

3.5. Dashboard and Results Visualization

- FR13: The system shall provide a dashboard displaying registered construction sites and their monitoring status.
- FR14: The system shall display satellite images, detected changes, and calculated change measurements.
- FR15: The system shall present monitoring results and detected changes through appropriate visual overlays.

3.7. Construction Site Registration and Management

- FR16: The system shall allow users to register a construction site by providing its name, geographical bounding-box coordinates, and construction start date.
- FR17: The system shall validate the provided coordinates and display the selected location on a map for confirmation before completing registration.

- FR18: The system shall allow users to view the details of their registered construction sites.

3.8. Construction Site Access and Member Management

1. FR19: The system shall allow users to request access to a construction site by providing its unique site identifier.
2. FR20: The system shall allow the creator of a construction site to approve or reject access requests from other users.

3.9. Construction Progress Viewing

1. FR21: The system shall allow users to view calculated change measurements for a registered construction site over time.
2. FR22: The system shall allow users to compare monitoring results from different dates to observe changes in the site.
3. FR23: The system shall allow users to view calculated change measurements for the latest satellite comparison of a registered construction site.

3.10. Monitoring Report Generation

- FR24: The system shall allow users to generate reports for a selected construction site and time period.
- FR25: The system shall include relevant satellite images, change detection results, and monitoring measurements in the generated reports.
- FR26: The system shall allow users to download generated reports for record-keeping and review.

3.11. Error Notifications and Handling

1. FR27: The system shall notify users when suitable satellite imagery is unavailable for the selected site or date range.
2. FR28: The system shall inform users when a monitoring operation cannot proceed because of a processing or service error.
3. FR29: The system shall notify users when submitted construction site information is invalid or incomplete.
4. FR30: The system shall display an appropriate error message when a monitoring operation fails.
5. FR31: The system shall handle database connectivity failures and external service errors that prevent required application operations from completing.

Note: All registered users will have the same access permissions and functional capabilities within the application. The functional requirements listed above

describe the system's expected behavior from the user's perspective. These requirements will be further refined into detailed System Functional Requirements (SFRs) to specify the system's operations and expected responses.

3.3 System Functional Requirements

The following detailed system-level requirements are retained from the supplied report. Each requirement table is preserved with its function description, inputs and origin, outputs and destination, dependencies, actions, preconditions, postconditions, and side effects where those fields were present.

1. User Account Management

This category covers account registration, login, and logout. We'll begin with three system-level requirements.

Requirement # 1: User Registration

Field	Description
Function Description	The system shall allow a new user to create an account by providing the required registration information.
Input & Origin	User registration information, including name, email address, and password, entered through the registration interface.
Outputs and Destination	A confirmation message indicating successful registration, displayed to the user. The newly created account information is stored in the database. If registration fails, an appropriate error message is displayed.
Requires	Registration interface, user authentication component, and database connectivity.
Actions to be taken	1. Receive the registration information from the user. 2. Validate that all required fields are provided and correctly formatted. 3. Check whether the email address is already associated with an existing account. 4. If the information is valid and the email is not registered, securely process the password and create the account. 5. Display the appropriate success or error message.
Pre-condition	The user is not logged in and has access to the registration interface.
Post-condition	Upon successful registration, a new user account exists in the database and can be used to access the system.
Side Effects	A new user record is created in the database. No construction site or monitoring record is created at this stage.

Requirement # 2: User Login

Field	Description
Function Description	The system shall allow registered users to log in using their valid email address and password.
Input & Origin	Email address and password entered by the user through the login interface.
Outputs and Destination	Upon successful authentication, the user is redirected to the main dashboard. If authentication fails, an appropriate error message is displayed.
Requires	Login interface, user authentication component, and database connectivity.
Actions to be taken	1. Receive the email address and password. 2. Validate that the required fields have been provided. 3. Retrieve the corresponding account record using the email address. 4. Verify the submitted password against the securely stored password information. 5. If the credentials are valid, establish an authenticated session or equivalent authentication state. 6. Redirect the user to the dashboard. Otherwise, display an error message.
Pre-condition	The user has an existing account and is not currently logged in.
Post-condition	Upon successful login, the user is authenticated and can access the system's available features.
Side Effects	An authenticated session or equivalent authentication state is established. No construction site data is modified by the login operation.

Requirement # 3: User Logout

Field	Description
Function Description	The system shall allow an authenticated user to log out of their account.
Input & Origin	A logout request initiated by the authenticated user through the application interface.
Outputs and Destination	A logout confirmation or redirection to the login page is provided to the user.
Requires	An active authenticated session or equivalent authentication state and the logout functionality.
Actions to be taken	1. Receive the logout request. 2. Invalidate the current authenticated session or authentication state. 3. Clear relevant authentication information maintained by the application. 4. Redirect the user to the login page or another publicly accessible page. 5. Ensure that protected pages and operations cannot be accessed using the invalidated authentication state.

Pre-condition	The user is logged in to the system.
Post-condition	The user's current authenticated session is terminated, and access to protected features requires authentication again.
Side Effects	The user may need to log in again to access protected features. Existing construction site records and monitoring results remain unchanged.

2. Data and Records Management

This category defines how the system stores, retrieves, and maintains records related to users, registered construction sites, satellite imagery, monitoring operations, and analysis results.

Requirement # 4: Store and Retrieve System Records

Field	**Description**
Function Description	The system shall store and retrieve the information required to operate the construction monitoring system.
Input & Origin	User account information from the registration process; construction site details from the site registration interface; satellite image metadata and references from GEE; monitoring results from the computer vision and AI processing pipeline.
Outputs and Destination	Requested records are retrieved from the database and provided to the relevant application component for display or processing. New records are stored in the database.
Requires	Database, database access component, and the relevant application modules.
Actions to be taken	1. Receive data from the relevant application component. 2. Validate the required data fields and formats. 3. Store new records in the appropriate database tables. 4. Retrieve requested records when needed by the application. 5. Return the retrieved records to the requesting component. 6. Report an appropriate error if a database operation fails.
Pre-condition	The database is configured and accessible, and the requesting application component has supplied the required information for the operation.
Post-condition	Valid information is stored successfully or the requested records are returned to the requesting component.
Side Effects	Database records may be created, updated, or read. A failed operation may leave the requested data unavailable until the problem is resolved.

Requirement # 5: Maintain Monitoring Operation Records

Field	Description
Function Description	The system shall maintain a record of each monitoring operation and its associated processing status and results.
Input & Origin	Monitoring operation details from the monitoring process management component, including the associated site, processing dates, operation status, and available result references.
Outputs and Destination	A monitoring record containing the operation details and current status is stored in the database and made available to the relevant application components.
Requires	Database, monitoring process management component, and computer vision processing pipeline.
Actions to be taken	1. Receive the details of a monitoring operation. 2. Associate the operation with the relevant registered construction site. 3. Create a monitoring record when the operation begins. 4. Update the record as processing progresses. 5. Store the final status and references to the generated results when processing finishes. 6. Record a failure status if the operation cannot be completed.
Pre-condition	A construction site has been registered, and a monitoring operation has been initiated or scheduled.
Post-condition	A record exists for the monitoring operation, reflecting its latest known status and, when available, its results.
Side Effects	A new monitoring record may be created or an existing record updated. The operation record itself does not guarantee that suitable satellite imagery or successful analysis results are available.

Requirement # 6: Associate Records with Their Corresponding Construction Sites

Field	Description
Function Description	The system shall associate satellite imagery metadata, monitoring operations, and analysis results with their corresponding registered construction sites.
Input & Origin	Construction site identifier, satellite image references and metadata from the imagery retrieval component, and monitoring result references from the processing pipeline.
Outputs and Destination	Associated records are stored in the database and can be retrieved by the relevant application components using the construction site identifier.

Requires	Database, construction site records, imagery retrieval component, and monitoring process management component.
Actions to be taken	1. Receive the construction site identifier with the relevant imagery, monitoring, or result information. 2. Verify that the referenced construction site exists. 3. Associate the information with that site using the appropriate database relationship. 4. Store the association and related information. 5. Reject the association and report an error if the site identifier is invalid.
Pre-condition	The construction site has been registered, and the relevant imagery, monitoring, or result information is available for association.
Post-condition	The relevant records are linked to the correct construction site and can be retrieved for site monitoring only.
Side Effects	Database relationships may be created or updated. No satellite image is downloaded or processed solely because a record is associated with a site.

3. Construction Change Detection

This category defines how the system retrieves suitable Sentinel-2 satellite imagery through Google Earth Engine (GEE), prepares images for comparison, detects changes between different dates, and presents the detected areas.

Requirement # 7: Retrieve Suitable Satellite Images

Field	**Description**
Function Description	The system shall retrieve suitable satellite imagery for a registered construction site and a specified monitoring period using Google Earth Engine.
Input & Origin	Construction site bounding-box coordinates and monitoring date range from the database; satellite imagery availability and metadata from GEE.
Outputs and Destination	References to suitable satellite images, their acquisition dates, and relevant metadata are provided to the computer vision processing pipeline and stored in the database. If no suitable imagery is available, the system reports this condition to the monitoring process management component.
Requires	Google Earth Engine access, registered construction site information, database connectivity, and an imagery retrieval component.

Actions to be taken	1. Retrieve the selected site's coordinates and requested date range. 2. Query GEE for Sentinel-2 imagery covering the site and period. 3. Apply appropriate date, spatial coverage, and image-quality filters, including cloud-related filtering. 4. Identify imagery suitable for comparison. 5. Return the image references and relevant metadata to the processing pipeline. 6. Store the references and metadata in the database. 7. Report when no suitable imagery is found or retrieval fails.
Pre-condition	A construction site has been registered, a valid monitoring period is available, and GEE access is configured.
Post-condition	Suitable image references and metadata are available for processing, or the system records that suitable imagery could not be retrieved.
Side Effects	GEE requests may consume available service quota or processing resources. Imagery retrieval does not guarantee that every requested date will have a usable image.

Requirement # 8: Preprocess Satellite Images for Comparison

Field	**Description**
Function	The system shall prepare retrieved satellite images to make them suitable for change detection.
Description	
Input & Origin	Satellite image pixel data, acquisition dates, and image metadata retrieved through GEE or made available to the processing environment.
Outputs and Destination	Preprocessed and comparable images are provided to the computer vision change detection component. Relevant preprocessing information may be stored with the monitoring record.
Requires	Satellite imagery retrieval component, image preprocessing component, and access to the actual image pixel data.
Actions to be taken	1. Retrieve or access the actual pixel data for the selected images. 2. Apply appropriate quality and cloud masking where required. 3. Ensure that images cover the registered construction site. 4. Prepare images using consistent spatial coverage, resolution, and data formats. 5. Check whether the resulting images are suitable for comparison. 6. Pass suitable image pairs to the change detection component or report why preprocessing could not be completed.
Pre-condition	At least two candidate satellite images are available for the selected site and monitoring period.
Post-condition	Suitable preprocessed image pairs are available for change detection, or the system identifies that the images cannot be compared reliably.

Side Effects	Temporary image data and intermediate processing files may be created. Preprocessing may exclude images or areas that do not meet the required quality conditions.

Requirement # 9: Detect Changes Between Satellite Images

Field	**Description**
Function Description	The system shall compare suitable satellite images acquired at different times and identify areas where detectable changes have occurred within a registered construction site.
Input & Origin	A suitable pair of preprocessed satellite images from the image preprocessing component and the corresponding site information from the database.
Outputs and Destination	A change mask identifying detected changed and unchanged areas, along with relevant processing metadata, is provided to the results visualization and analysis components.
Requires	Preprocessed image pairs, a trained or configured computer vision change detection model, and the processing environment.
Actions to be taken	1. Receive the selected image pair and associated metadata. 2. Provide the images to the change detection model. 3. Compare the images to identify areas exhibiting detectable changes. 4. Generate a change mask representing the detected changes. 5. Associate the output with the corresponding site and image dates. 6. Pass the change mask and processing information to the relevant components. 7. Report an error if inference fails.
Pre-condition	Two suitable, preprocessed satellite images of the same site from different dates are available, and the change detection model is ready for use.
Post-condition	A change mask and associated metadata are generated and made available to the results processing components, or a failure status is recorded.
Side Effects	Processing consumes computing resources. Detection errors may occur because of image quality, resolution, seasonal differences, shadows, or model limitations.

Requirement # 10: Calculate and Store Detected Change Measurements

Field	**Description**
Function Description	The system shall calculate the estimated area represented by detected changes and associate the measurements with the corresponding monitoring operation.

Input & Origin	Generated change mask, image resolution or spatial metadata, site boundary information, and monitoring dates from the change detection component and database.
Outputs and Destination	Estimated changed-area measurements, measurement units, and associated dates are stored in the database and provided to the dashboard and AI-based analysis components.
Requires	Change mask, geospatial metadata, area calculation component, and database connectivity.
Actions to be taken	1. Receive the generated change mask and relevant spatial metadata. 2. Identify the pixels classified as changed within the site boundary. 3. Calculate the estimated area represented by the detected pixels using the appropriate spatial resolution or geospatial calculation method. 4. Associate the measurement with the relevant image dates and monitoring operation. 5. Store the measurement and its unit in the database. 6. Report an error if the required spatial information is unavailable or the calculation fails.
Pre-condition	A valid change mask and sufficient spatial metadata are available for the registered construction site.
Post-condition	The estimated changed area is calculated and stored with its corresponding monitoring record, or the system records that the measurement could not be calculated.
Side Effects	New measurement records may be created. The calculated area represents detected image changes and does not necessarily equal the actual construction area or percentage of construction completed.

Requirement # 11: Display Change Detection Results

Field	**Description**
Function Description	The system shall display the satellite images used for comparison and the corresponding detected changes to the user.
Input & Origin	Original or processed satellite images, acquisition dates, generated change masks, and calculated change measurements from the database and processing pipeline.
Outputs and Destination	A visual comparison of the selected images, highlighted changed areas, acquisition dates, and available change measurements is displayed on the application's results interface.
Requires	Results visualization component, image and result references, and access to the corresponding image data and generated outputs.

Actions to be taken	1. Receive the selected monitoring result. 2. Retrieve the corresponding satellite images or their stored representations. 3. Retrieve the associated change mask and measurements. 4. Display the images with their acquisition dates. 5. Overlay or otherwise visualize the detected changed areas. 6. Display the available measurements and report an error if the required results cannot be retrieved.
Pre-condition	At least one completed change detection operation has produced results available for display.
Post-condition	The user can view the relevant images, detected changes, and available measurements for the selected monitoring operation.
Side Effects	Image and result data may be loaded into the user's browser or application session. Displaying results does not modify the stored analysis records.

4. AI-Based Construction Analysis

This category defines how the system interprets the outputs of computer vision analysis and presents understandable observations about changes detected at a construction site. AI-generated insights must be based on available monitoring results and should not claim more than the satellite imagery supports.

Requirement # 12: Analyze Detected Construction Changes

Field	**Description**
Function Description	The system shall analyze detected change measurements and associated monitoring information to identify patterns in construction site development over time.
Input & Origin	Change masks, estimated changed areas, satellite image acquisition dates, and historical monitoring results from the computer vision pipeline and database.
Outputs and Destination	Analysis findings, identified change patterns, and relevant observations are provided to the dashboard and AI-generated insights component.
Requires	Completed change detection results, stored monitoring records, and an analysis component.
Actions to be taken	1. Retrieve the available change measurements and corresponding monitoring dates. 2. Identify observable patterns, such as increases or decreases in detected changed areas. 3. Generate analysis findings based on the available evidence. 4. Make the findings available to the dashboard and insights component.
Pre-condition	At least one completed monitoring operation has produced valid change measurements.

Post-condition	Analysis findings are generated from the available monitoring results and associated with the relevant construction site and monitoring period.
Side Effects	Additional analysis records may be stored. Incomplete or inconsistent monitoring data may limit the reliability of identified patterns.

Requirement # 13: Generate AI-Based Construction Insights

Field	**Description**
Function	The system shall generate understandable AI-based observations
Description	from available construction monitoring results to help users interpret detected changes.
Input & Origin	Change detection measurements, image dates, and relevant analysis findings supplied by the computer vision pipeline and database.
Outputs and Destination	A textual summary of observed changes, possible development patterns, and relevant limitations is provided to the dashboard.
Requires	Available monitoring results, analysis findings, and an AI-based insight generation component.
Actions to be taken	1. Retrieve the relevant monitoring results and analysis findings. 2. Prepare the available measurements, and dates as input to the AI component. 3. Generate a concise explanation of the observed changes. 4. Ensure that the explanation is based on the supplied evidence and does not present unsupported conclusions as facts. 5. Associate the generated insights with the corresponding site and monitoring period. 6. Provide the insights to the dashboard. 7. Report an appropriate error if insight generation fails.
Pre-condition	Relevant monitoring results or analysis findings are available, and the AI component is configured and accessible.
Post-condition	AI-generated observations are available for display, or the system records that insight generation was unsuccessful.
Side Effects	AI processing may consume computing resources or external service quota, depending on the implementation. Generated observations may contain inaccuracies and must not be treated as definitive evidence of construction completion or structural safety.

Requirement # 14: Display AI Analysis and Its Supporting Measurements

Field	Description
Function Description	The system shall display AI-generated construction insights alongside the corresponding monitoring measurements and dates.
Input & Origin	Generated AI observations, estimated changed-area measurements, monitoring dates, and for a selected monitoring check and allow users to generate and download reports. from the analysis component and database.
Outputs and Destination	An integrated presentation of AI insights and supporting monitoring data is displayed on the construction site's results dashboard.
Requires	AI insight generation component, stored monitoring results, database connectivity, and dashboard visualization component.
Actions to be taken	1. Retrieve the AI-generated observations for the selected site or monitoring period. 2. Retrieve the measurements and dates associated with those observations. 3. Display the observations alongside their supporting information. 4. Identify the relevant monitoring period for each observation. 5. Indicate when AI insights are unavailable or could not be generated.
Pre-condition	The user has selected a registered construction site or monitoring record, and the relevant results are available.
Post-condition	The user can review the generated observations together with the measurements and monitoring dates used to support them.
Side Effects	Displaying the analysis does not modify the underlying satellite imagery, change masks, or monitoring measurements.

5. Dashboard and Results Visualization

This category defines how the system presents registered construction sites, monitoring operations, satellite imagery, detected changes, measurements, and AI-generated insights through a user-friendly dashboard.

Requirement # 15: Display the Main Dashboard

Field	Description
Function Description	The system shall provide an interactive dashboard displaying the user's registered construction sites and their monitoring status.

Input & Origin	Registered construction site details and monitoring status records retrieved from the database.
Outputs and Destination	A dashboard containing the registered site list, site details, and monitoring statuses is displayed to the user.
Requires	User authentication, database connectivity, construction site management component, and dashboard interface.
Actions to be taken	1. Verify that the user is authenticated. 2. Retrieve the construction sites associated with the user's account. 3. Retrieve the latest available monitoring status for each site. 4. Display the site names and relevant summary information. 5. Allow the user to select a site to view its details and monitoring results. 6. Display an appropriate message if no sites have been registered or if the information cannot be retrieved.
Pre-condition	The user is authenticated and the dashboard interface is accessible.
Post-condition	The dashboard displays the available construction site information and monitoring statuses, or an appropriate empty-state or error message.
Side Effects	Dashboard data may be retrieved from the database. No site details or monitoring results are modified by displaying the dashboard.

Requirement # 16: Display Construction Site Monitoring Results

Field	**Description**
Function Description	The system shall display the available monitoring results for a selected construction site.
Input & Origin	Selected construction site identifier, satellite image references, acquisition dates, change masks, estimated changed areas, and AI-generated insights from the database and processing components.
Outputs and Destination	A site-specific results view displaying available satellite images, detected changes, measurements, monitoring dates, and AI-generated observations.
Requires	Registered construction site information, completed monitoring results, database connectivity, and results visualization component.
Actions to be taken	1. Receive the user's selection of a construction site. 2. Retrieve the associated monitoring records. 3. Retrieve the available image and change detection output references. 4. Display the relevant images, acquisition dates, and detected changes. 5. Display calculated measurements and available AI-generated insights. 6. Indicate when results are unavailable or could not be retrieved.

Pre-condition	The user is authenticated and has selected a registered construction site.
Post-condition	The available monitoring results for the selected site are displayed, or the system informs the user that no results are available.
Side Effects	Relevant images and result data may be loaded into the application session. Stored monitoring records remain unchanged.

Requirement # 17: Display Construction Site Location on a Map

Field	**Description**
Function Description	The system shall display the geographical location and registered bounding box of a selected construction site on a map.
Input & Origin	Construction site bounding-box coordinates and site name retrieved from the database.
Outputs and Destination	A map displaying the selected construction site's location and geographical boundary is presented to the user.
Requires	Registered site coordinates, map visualization component, and access to the map data or map service used by the application.
Actions to be taken	1. Receive the selected construction site identifier. 2. Retrieve its stored coordinates. 3. Convert the coordinates into the format required by the map component. 4. Display the location and bounding box on the map. 5. Adjust the map view to show the selected site. 6. Display an appropriate message if the coordinates are unavailable or cannot be displayed.
Pre-condition	The user is authenticated and has selected a registered construction site with valid geographical coordinates.
Post-condition	The site's registered location and boundary are displayed on the map, or the system reports that the location cannot be displayed.
Side Effects	The map component may request map data from an external service. Displaying the location does not modify the registered coordinates.

7. Construction Site Registration and Management

This category defines how users register construction sites, validate geographical coordinates, confirm the selected location on a map, and view their registered site information.

Requirement # 18: Register a Construction Site

Field	Description
Function Description	The system shall allow an authenticated user to register a construction site by providing its name, geographical bounding-box coordinates, and construction start date.
Input & Origin	Site name, four bounding-box coordinates (minimum and maximum latitude and longitude), and construction start date entered through the site registration interface.
Outputs and Destination	Upon successful registration, a confirmation message is displayed, and the construction site information is stored in the database. If registration fails, an appropriate error message is displayed.
Requires	User authentication, site registration interface, coordinate validation component, database connectivity, and map visualization component.
Actions to be taken	1. Receive the site name, bounding-box coordinates, and construction start date. 2. Validate that all required fields are provided. 3. Validate the coordinate values and ensure that the bounding box is geographically valid. 4. Verify that the construction start date is valid and not in the future. 5. Display the selected location on a map for user confirmation. 6. After confirmation, store the site information in the database and associate it with the registering user. 7. Display a success message or an appropriate validation or storage error.
Pre-condition	The user is authenticated and has access to the construction site registration interface.
Post-condition	A valid, confirmed construction site is stored in the database and associated with the registering user, or registration remains incomplete if validation, confirmation, or storage fails.
Side Effects	A new construction site record may be created. The site becomes available for subsequent monitoring operations after successful registration.

Requirement # 19: Validate and Confirm Construction Site Coordinates

Field	Description
Function Description	The system shall validate the submitted geographical coordinates and display the corresponding bounding box on a map before completing site registration.
Input & Origin	Minimum and maximum latitude and longitude entered by the user through the site registration interface.

Outputs and Destination	Coordinate validation results and a map displaying the selected geographical boundary are presented to the user. Invalid coordinates result in an appropriate error message.
Requires	Site registration interface, coordinate validation component, and map visualization component.
Actions to be taken	1. Receive the four bounding-box coordinates. 2. Check that all coordinate values are numeric and within valid latitude and longitude ranges. 3. Verify that the minimum latitude is less than the maximum latitude and the minimum longitude is less than the maximum longitude. 4. If the coordinates are valid, display the bounding box on the map. 5. Ask the user to confirm that the displayed location is correct. 6. Allow the user to correct the coordinates if the location is incorrect. 7. Prevent registration from proceeding with invalid or unconfirmed coordinates.
Pre-condition	The user has opened the site registration interface and entered the geographical coordinates.
Post-condition	Valid coordinates are displayed for confirmation, or the user is informed that the coordinates are invalid. Site registration can proceed only after successful validation and confirmation.
Side Effects	The map component may request map data from an external service. No construction site record is finalized solely by validating or displaying the coordinates.

Requirement # 20: View Registered Construction Site Details

Field	**Description**
Function Description	The system shall allow authenticated users to view the details of their registered construction sites.
Input & Origin	A selected construction site identifier or a request to view registered sites submitted through the dashboard.
Outputs and Destination	The site's name, geographical coordinates, construction start date, and other available site information are displayed to the user.
Requires	User authentication, construction site records, database connectivity, and site details interface.

Actions to be taken	1. Receive the request to view registered sites or a specific site's details. 2. Retrieve the relevant site records associated with the user's account. 3. If a specific site is selected, retrieve its stored details. 4. Display the available information in the site details interface. 5. Display an appropriate message if no sites are registered or the requested record cannot be retrieved.
Pre-condition	The user is authenticated and has access to the site management interface.
Post-condition	The requested registered site information is displayed, or the system informs the user that the information is unavailable.
Side Effects	Site records may be retrieved from the database. Viewing site details does not modify the stored information.

8. Construction Site Registration and Management

Requirement # 21: Request Access to a Construction Site

Field	Description
Function Description	The system shall allow users to request access to a registered construction site by providing its unique site identifier.
Input & Origin	Unique construction site identifier entered by the user and validated against construction site records stored in the database.
Outputs and Destination	A confirmation message indicating successful submission, or an appropriate message explaining why the request could not be submitted, shall be displayed to the user. The request details shall be stored in the database for review by the construction site creator.
Requires	User authentication and an existing registered construction site.
Actions to be Taken	Verify that the site exists and check whether the user already has access or has an existing pending request. If no duplicate request exists, create an access request associated with the user and the construction site.
Pre-condition	The user must be logged in, and the construction site must be registered in the system.
Post-condition	A valid access request is stored with a pending status for review by the construction site creator.
Side Effects	A new access request record is added to the database.

Requirement # 22: Approve or Reject Construction Site Access Requests

Field	Description
Function Description	The system shall allow the creator of a construction site to review access requests and approve or reject requests submitted by other users.
Input & Origin	Construction site identifier, access request details, and the approval or rejection decision selected by the construction site creator through the user interface.
Outputs and Destination	The updated request status and the result of the decision shall be displayed to the construction site creator. The access request status shall be updated in the database, and access permissions shall be updated when the request is approved.
Requires	User authentication, construction site records, and pending access requests.
Actions to be Taken	Verify that the user making the decision is the creator of the relevant construction site and that the request is pending. If approved, grant the requesting user access and update the request status to approved. If rejected, update the request status to rejected without granting access.
Pre-condition	The user must be logged in as the creator of the relevant construction site, and the access request must have a pending status.
Post-condition	The access request status is updated to approved or rejected. Access is granted only when the request is approved.
Side Effects	The access request record and, when applicable, the site's membership and access permissions are updated in the database.

8. Construction Progress Monitoring

This category defines how the system presents monitoring results over time, compares observations from different dates, and helps users assess changes in construction sites using available satellite imagery and calculated measurements.

Requirement # 23: View Construction Change Measurements Over Time

Field	**Description**
Function Description	The system shall allow users to view calculated changed-area measurements for a registered construction site viewing the latest calculated change measurement.
Input & Origin	Selected construction site identifier and available monitoring records containing acquisition dates and estimated changed-area measurements, retrieved from the database.

Outputs and Destination	A chronological list or visualization of available changed-area measurements, including their corresponding monitoring dates and measurement units, is displayed to the user.
Requires	Registered construction site information, historical monitoring records, database connectivity, and dashboard visualization component.
Actions to be taken	1. Receive the selected construction site identifier. 2. Retrieve the site's available monitoring measurements and associated dates. 3. Arrange the measurements in chronological order. 4. Display each measurement with its corresponding date and unit. 5. Indicate when measurements are unavailable for some monitoring periods. 6. Display an appropriate message if no valid measurements are available.
Pre-condition	The user is authenticated and has selected a registered construction site.
Post-condition	Available change measurements are displayed chronologically, or the system informs the user that no valid measurements are available.
Side Effects	Historical monitoring records may be retrieved for display. Stored measurements remain unchanged.

9. Monitoring Report Generation

This category defines how users generate and download reports containing construction site monitoring information, satellite imagery, detected changes, measurements, and relevant AI-generated insights.

Requirement # 24: Generate a Construction Monitoring Report

Field	**Description**
Function Description	The system shall allow users to generate a monitoring report for a selected construction site and monitoring period.
Input & Origin	Selected construction site, requested date range, and report generation request submitted by the user through the dashboard.
Outputs and Destination	A monitoring report containing the available information for the selected site and period is generated and made available to the user.
Requires	Registered construction site information, monitoring records, report generation component, database connectivity, and access to the relevant monitoring results.

Actions to be taken	1. Receive the selected construction site and requested date range. 2. Validate the selected site and date range. 3. Retrieve the relevant monitoring records within the requested period. 4. Collect the available change detection results, measurements, and AI-generated insights. 5. Organize the information into a structured report. 6. Generate the report in pdf. 7. Make the generated report available for download. 8. Display an appropriate message if the report cannot be generated or the requested period contains no usable results.
Pre-condition	The user is authenticated, a construction site is selected, and the requested reporting period is valid.
Post-condition	A report is generated for the selected site and period when sufficient information is available, or the system informs the user that report generation was unsuccessful.
Side Effects	Report files may be created in temporary or persistent storage. Report generation may consume processing resources.

Requirement # 25: Include Monitoring Information in the Report

Field	**Description**
Function Description	The system shall include the relevant construction site and monitoring information available for the selected reporting period in the generated report.
Input & Origin	Construction site details, monitoring dates, estimated changed-area measurements, and AI-generated insights retrieved from the database and processing pipeline.
Outputs and Destination	A structured report containing the applicable site information, monitoring observations, and supporting visual or numerical results is provided to the report generation component.
Requires	Site records, completed monitoring results, image and output references, report generation component, and database connectivity.

Actions to be taken	1. Retrieve the registered construction site details. 2. Retrieve monitoring records corresponding to the selected period. 3. Include the site name, geographical information, and reporting period. 4. Include acquisition dates. 5. Include the corresponding change detection visualizations and estimated measurements. 6. Include available AI-generated observations and relevant limitations. 7. Ensure that each result is associated with its corresponding monitoring date. 8. Pass the assembled information to the report generation component.
Pre-condition	A valid report request exists, and relevant site information and monitoring records are available.
Post-condition	The report contains the available information relevant to the selected site and reporting period, with results associated with their corresponding dates.
Side Effects	Selected images and analysis results may be retrieved or temporarily assembled for report generation. The original records remain unchanged.

Requirement # 26: Download a Generated Monitoring Report

Field	**Description**
Function	The system shall allow users to download a successfully
Description	generated construction monitoring report.
Input &	A download request for a generated report, initiated by the user
Origin	through the report interface.
Outputs and	The generated report file is delivered to the user's browser for
Destination	downloading to their device.
Requires	A successfully generated report, report file storage or retrieval mechanism, user authentication, and report download functionality.
Actions to be taken	1. Receive the user's download request. 2. Identify the requested report. 3. Create a Report 4. Deliver the file to the user's browser in pdf format. 6. Display an appropriate error message if the report is unavailable or the download fails.
Pre-condition	The user is authenticated, and a successfully generated report is available for download.

Post-condition	The report file is made available to the user for downloading, or the system informs the user that the download could not be completed.
Side Effects	Network bandwidth may be consumed during the download. Downloading a report does not modify the stored monitoring results.

10. Error Notifications and Handling

This category defines how the system identifies errors, informs users about problems, and handles situations that prevent site registration, satellite imagery retrieval, monitoring, analysis, or report generation from completing successfully.

Requirement # 27: Notify Users About Invalid or Incomplete Site Information

Field	**Description**
Function Description	The system shall notify users when submitted construction site information is incomplete or invalid.
Input & Origin	Site name, geographical bounding-box coordinates, construction start date, and other required site information submitted through the site registration interface.
Outputs and Destination	An appropriate validation error message is displayed to the user, identifying the field or information that requires correction.
Requires	Site registration interface, input validation component, and coordinate validation component.
Actions to be taken	1. Receive the submitted construction site information. 2. Check that all required fields are provided. 3. Validate the field formats and coordinate ranges. 4. Verify that the bounding-box coordinates define a valid geographical area. 5. Validate the construction start date. 6. Identify any invalid or missing information. 7. Display an appropriate error message and prevent registration until the required corrections are made.
Pre-condition	The user has opened the site registration interface and submitted or attempted to submit construction site information.
Post-condition	Invalid or incomplete information is identified and communicated to the user. The site is not registered until the required validation checks are passed.
Side Effects	Validation messages may be displayed, but no valid construction site record is created from the rejected submission.

Requirement # 28: Notify Users When Suitable Satellite Imagery Is Unavailable

Field	Description
Function Description	The system shall notify users when suitable satellite imagery cannot be found for the selected construction site and monitoring period.
Input & Origin	Construction site coordinates, requested monitoring date range, and imagery availability or quality information returned by GEE.
Outputs and Destination	A notification explaining that suitable satellite imagery is unavailable is displayed to the user and associated with the relevant monitoring operation.
Requires	GEE access, imagery retrieval component, image-quality filtering component, and monitoring process management component.
Actions to be taken	1. Receive the imagery retrieval results. 2. Determine whether suitable images are available for the requested site and period. 3. Identify whether the issue results from unavailable imagery, image quality, cloud cover, or insufficient suitable image pairs, where this information is available. 4. Record the appropriate monitoring status. 5. Display an understandable notification to the user. 6. Allow the user to select another monitoring period or retry when appropriate.
Pre-condition	A monitoring operation has requested satellite imagery for a registered construction site.
Post-condition	The user is informed when suitable imagery is unavailable, and the monitoring operation is recorded with an appropriate status.
Side Effects	The monitoring operation may remain incomplete. Previously completed monitoring results remain available.

Requirement # 29: Handle Monitoring and Processing Errors

Field	Description
Function Description	The system shall detect and handle errors occurring during satellite imagery retrieval, image preprocessing, change detection, measurement calculation, or AI analysis.
Input & Origin	Error codes, exception information, service responses, and processing status information generated by the relevant system components.
Outputs and Destination	An appropriate error notification is displayed to the user, and relevant failure information is recorded in the monitoring operation record.
Requires	Error handling mechanism, monitoring process management component, processing pipeline, database connectivity, and logging facility.

Actions to be taken	1. Receive an error from the affected component. 2. Identify the associated monitoring operation and processing stage. 3. Record the failure status and relevant diagnostic information. 4. Prevent the operation from being incorrectly marked as successfully completed. 5. Preserve previously completed results. 6. Display a user-friendly notification without exposing sensitive technical details. 7. Permit a retry when appropriate and safe.
Pre-condition	A monitoring or analysis operation is in progress, and a system component reports an error.
Post-condition	The error is handled, relevant failure information is recorded, and the user is informed of the unsuccessful operation or affected processing stage.
Side Effects	Partial outputs or temporary processing files may remain and require cleanup. Retrying an operation may consume additional computing resources or GEE service quota.

Requirement # 30: Handle Report Generation and Download Errors

Field	**Description**
Function Description	The system shall notify users when a monitoring report cannot be generated or downloaded successfully.
Input & Origin	Report generation requests, report data availability information, file storage responses, and download status information from the report generation and download components.
Outputs and Destination	An appropriate error message is displayed to the user, and relevant diagnostic information is recorded where applicable.
Requires	Report generation component, report storage or retrieval mechanism, database connectivity, and error handling mechanism.
Actions to be taken	1. Receive an error from the report generation or download component. 2. Identify the affected report operation. 3. Determine whether the failure occurred during data retrieval, report creation, file storage, or file delivery. 4. Record the operation's failure status and relevant diagnostic information. 5. Display an appropriate message explaining that the report is unavailable or the operation failed. 6. Allow the user to retry when appropriate and safe.
Pre-condition	A user has requested a monitoring report to be generated or downloaded.
Post-condition	The user is informed of the failure, and the system records the relevant operation status where applicable. If the failure is resolved and the operation is retried successfully, the report becomes available.

Side Effects	Incomplete report files or temporary files may require cleanup. A retry may consume additional processing resources.

Requirement # 31: Handle Database and External Service Errors

Field	**Description**
Function Description	The system shall handle database connectivity failures and external service errors that prevent required application operations from completing.
Input & Origin	Database error responses, GEE service errors, authentication or access failures, network errors, and other relevant service status information.
Outputs and Destination	An appropriate error status is recorded where possible, and a user-friendly notification is displayed when the failure affects a requested operation.
Requires	Database access component, external service integration components, error handling mechanism, and application logging facility.
Actions to be taken	1. Detect a database or external service failure. 2. Identify the affected operation and service. 3. Record the error details when logging or database access remains available. 4. Prevent incomplete operations from being reported as successful. 5. Display an appropriate message to the user. 6. Retry transient failures only when appropriate and safe. 7. Resume or restart the affected operation when the service becomes available and the operation can safely continue.
Pre-condition	An application operation requires database access or an external service, and the required resource becomes unavailable or returns an error.
Post-condition	The failure is handled and communicated appropriately. The operation either completes successfully after recovery or remains marked as failed or incomplete.
Side Effects	Some operations may be delayed or temporarily unavailable. Error logging may be incomplete if the database or logging service itself is unavailable. Repeated retries may consume additional resources.

Requirement # 32: Handle Monitoring Operation Failures

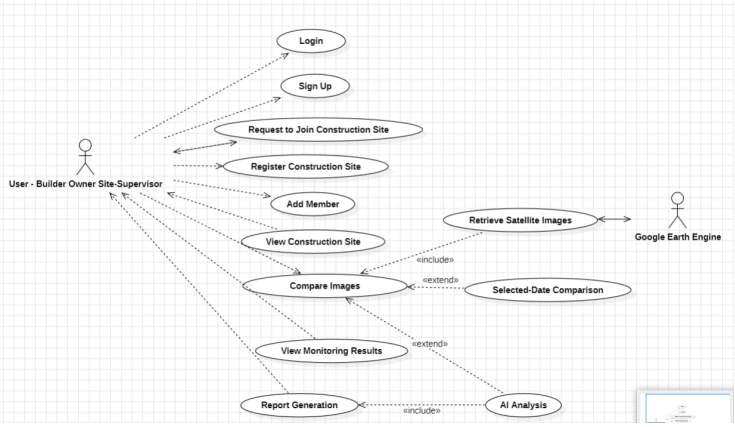
Field	**Description**
Function Description	The system shall identify failures that prevent a monitoring operation from completing successfully.

Input & Origin	Error information from GEE requests, image preprocessing, computer vision inference, measurement calculation, database operations, or other monitoring components.
Outputs and Destination	The system shall display the operation's failure status and details on the user interface.
Requires	Monitoring process management component, error handling mechanism, database connectivity, and access to operation status records.
Actions to be taken	1. Receive an error reported by a monitoring component. 2. Identify the affected monitoring operation.4. Prevent the failed operation from being incorrectly marked as successfully completed. 5. Display a user-appropriate error message. 6. Preserve previously completed monitoring results. 7. Allow a new attempt when the cause has been addressed and the operation can safely be retried.
Pre-condition	A monitoring operation has been initiated, and a component reports an error or failure.
Post-condition	The failure is communicated appropriately, while previously completed results remain available.
Side Effects	Partial processing outputs or temporary files may remain and may require cleanup. A retry may consume additional resources or external service quota.

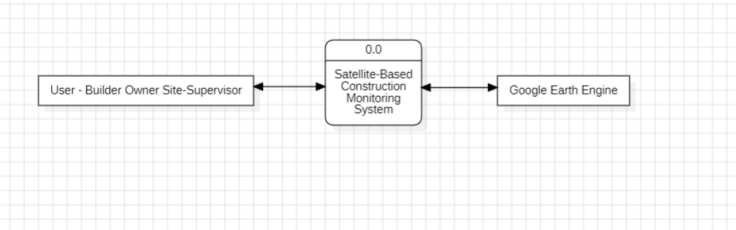
3.4 Behaviour Requirements

This section contains the interaction and behavioural models associated with the requirements above.

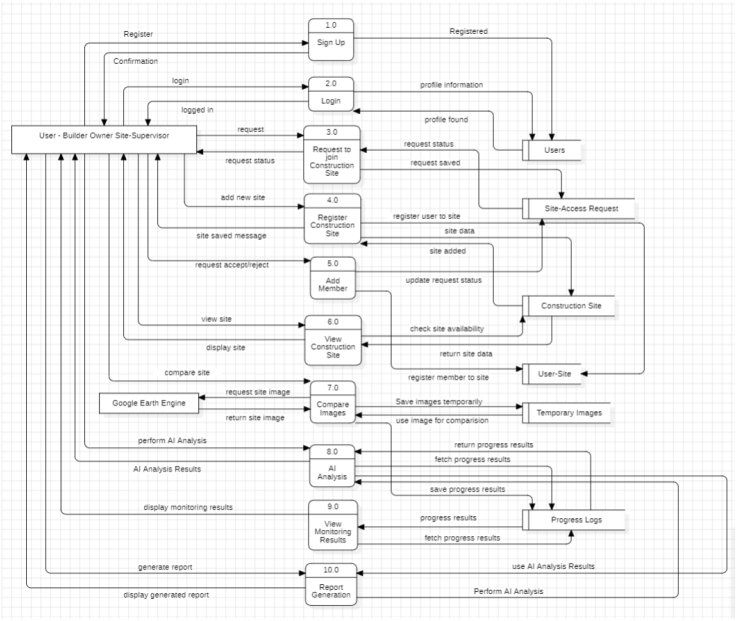
3.4.1 Use Case Diagram



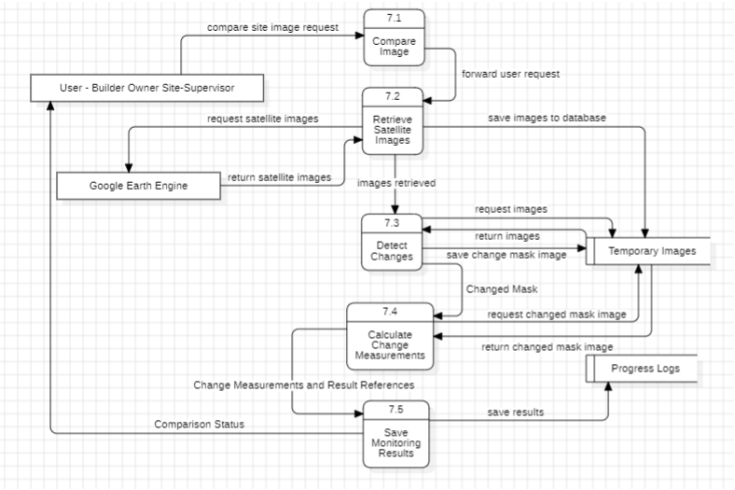
3.4.2 Data Flow Diagram — Level 0



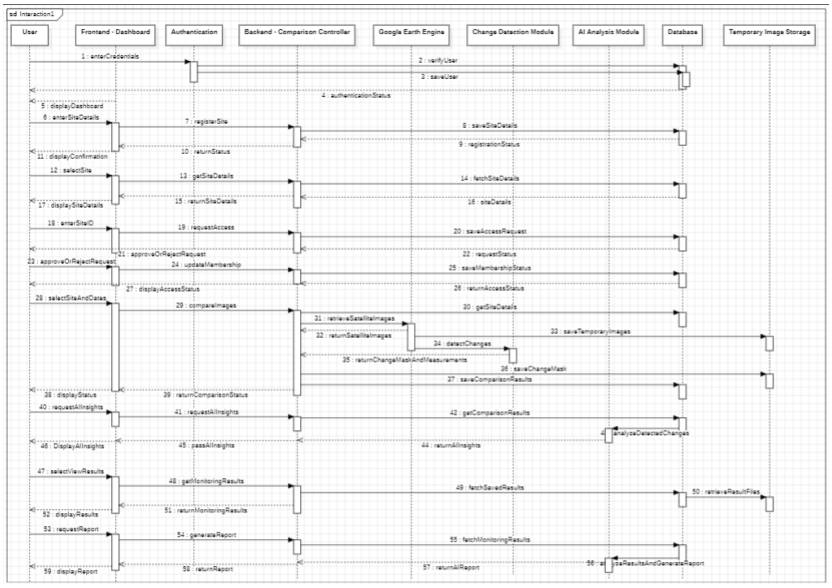
3.4.3 Data Flow Diagram — Level 1



3.4.4 Data Flow Diagram — Level 2: Compare Images



3.4.5 Sequence Diagram



4. Other Non-Functional Requirements

5. Non-Functional Requirements (NFRs)

1. Performance and Response Time

Req ID	Requirement Name	Non-Functional Requirement Specification
NFR-PERF-01	Dashboard Load Time	The system shall load and display registered construction sites within 3 seconds.
NFR-PERF-02	Image Processing Time	The system shall complete satellite image preprocessing and change detection within 120 seconds per bounding box for the defined test area, number of image pairs, and hardware setup. If processing takes longer, it shall continue in the background and show the current status to the user.
NFR-PERF-03	Report Generation Time	The system shall generate a PDF monitoring report within 10 seconds, provided that the required monitoring data and images are already available.
NFR-PERF-04	User Interface Responsiveness	The system shall provide visible feedback within 1 seconds when a user performs an action, such as selecting a date, entering coordinates, or changing a visualization layer.

2. Accuracy and Quality

Req ID	Requirement Name	Non-Functional Requirement Specification
NFR-ACC-01	Change Detection Model Quality	The computer vision model shall achieve an Intersection over Union (IoU) score of at least 0.50 on a separate test dataset containing manually labelled satellite images representative of the project's target imagery.
NFR-ACC-02	Spatial Resolution	The system shall process selected Sentinel-2 bands at their native 10-metre spatial resolution, where applicable. It shall not claim that the detected changes provide more spatial detail than the source imagery supports.
NFR-ACC-03	Input Validation	The system shall reject all invalid inputs included in the defined validation test cases, including missing required fields, coordinates outside valid ranges, incorrect bounding boxes, and invalid construction start dates.

3. Availability and Reliability

Req ID	Requirement Name	Non-Functional Requirement Specification
NFR-REL-01	System Availability	The web application and backend API shall remain available for at least 95% of each calendar month, excluding scheduled maintenance.
NFR-REL-02	External Service Error Handling	If Google Earth Engine (GEE) or another external service fails, the system shall display an appropriate error message within 2 seconds of detecting the failure and record the status of the affected operation whenever possible.

4. Capacity and Scalability

Req ID	Requirement Name	Non-Functional Requirement Specification
NFR-SCA-01	Construction Site Capacity	The system shall support at least 1,000 registered construction sites without requiring changes to the database structure.
NFR-SCA-02	Processed Image Caching	The system shall temporarily store eligible processed image outputs for up to 10 days to reduce repeated processing and unnecessary requests to external services. Cached results shall be updated when their source data or processing settings change.

5. Security and Compliance

Req ID	Requirement Name	Non-Functional Requirement Specification
NFR-SEC-01	Session Inactivity Timeout	The system shall automatically end an authenticated session after 1 hour of inactivity. The user shall log in again to access protected features.
NFR-SEC-02	Password Length	The system shall require passwords to contain at least 8 characters and shall store them using Django's password-hashing method instead of storing them as plain text.

5. Other Requirements

No additional requirements beyond those specified in Chapters 2–4 and Appendix A have been identified at this stage.

Appendix A — Data Dictionary

Data Model and Database Specification:

Data Dictionary

1.1 users Table

This table stores the account information of registered users.

· **user_id** (UUID, Primary Key): A unique identifier assigned to each user.

· **email** (VARCHAR, Unique, Not Null): The user's email address, used for account identification.

· **password_hash** (VARCHAR, Not Null): The securely hashed password, if password management is handled by the application.

· **full_name** (VARCHAR): The full name of the user.

1.2 construction_sites Table

This table stores the details of construction sites registered for monitoring.

· **construction_id** (UUID, Primary Key): A unique identifier assigned to each construction site.

· **site_name** (VARCHAR, Not Null): The name of the construction site.

· **created_by** (UUID, Foreign Key → users.user_id): Identifies the user who registered the site.

· **bounding_boundaries** (JSON) The coordinates pointing to the boundaries of the whole construction site.
location (VARCHAR): A general address or description of the site's location.

· **city** (VARCHAR): The city in which the site is located.

· **country** (VARCHAR): The country in which the site is located.

· **area** (DECIMAL): The approximate area of the construction site in square metres.

· **start_date** (DATE, Not Null): The construction start date provided by the user.

· **is_completed** (BOOLEAN, Default: False): Indicates whether the user has marked the construction project as completed.

· **end_date** (DATE, Nullable): The actual completion date, if the project has been completed.

· **is_public** (BOOLEAN, Default: False): Indicates whether the site is publicly viewable.

· **created_at** (TIMESTAMP): The date and time when the site record was created.

· **current_progress** (DECIMAL): Latest calculated overall progress (0.00% to 100.00%).

1.3 user_site Table

This table links users to construction sites they are permitted to access. It supports a many-to-many relationship because a user may access multiple sites, and a site may be associated with multiple users.

· **User_site_id** (UUID, Primary Key): A unique identifier for the access record.

· **user_id** (UUID, Foreign Key → users.user_id): Identifies the user receiving access.

· **construction_id** (UUID, Foreign Key → construction_sites.construction_id): Identifies the construction site.

· **granted_at** (TIMESTAMP): The date and time when access was granted.

1.4 site_access_request Table

This table stores requests from users who want to access private construction sites.

· **request_id** (UUID, Primary Key): A unique identifier for each request.

· **request_from_id** (UUID, Foreign Key → users.user_id): Identifies the user requesting access.

· **request_to_id** (UUID, Foreign Key → users.user_id): Identifies the user receiving the request, usually the site's owner.

· **construction_id** (UUID, Foreign Key → construction_sites.construction_id): Identifies the site for which access is requested.

· **status** (VARCHAR, Default: 'PENDING'): Indicates whether the request is pending, approved, or rejected.

· **created_at** (TIMESTAMP): The date and time when the request was submitted.

1.5 progress_logs Table

This table stores the latest run output for the active site check. performed on a construction site. Each record represents a comparison between satellite images taken at different times.

1. **progress_id** (UUID, Primary Key): A unique identifier for each monitoring operation.
2. **construction_id** (UUID, Foreign Key → construction_sites.construction_id): Identifies the construction site being monitored.
3. **Baseline_image_file_name** (VARCHAR, Nullable): The filename of the original baseline satellite image stored in Supabase Storage (e.g., construction_id___baseline_date.tif).
4. **latest_image_filename (VARCHAR**, Nullable**): The filename of the latest satellite image stored in Supabase Storage using the standard format: <construction_id>___<date_of_satellite_image>.tif (e.g., c123_456___2026-10-19.tif).**
5. **change_mask_filename (VARCHAR**, Nullable**): The filename of the OpenCV-generated visual overlay mask stored in Supabase Storage (e.g., c123_456___2026-10-19_mask.tif).**
6. **area_changed_sqm** (DECIMAL, Nullable): The estimated area of detected change, in square metres.
7. **change_percentage** (DECIMAL, Nullable): The percentage of the monitored site area classified as changed, if calculated by the system.
8. **status** (VARCHAR): The current status of the monitoring operation, such as PENDING, PROCESSING, COMPLETED, or FAILED.
9. **Satellite_image_date** (DATE): Date the satellite image was captured.
10. **processed_at** (TIMESTAMP, Nullable): The date and time when the monitoring operation finished processing.

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2. ENTITY RELATIONSHIPS

The database tables are connected through primary and foreign keys to maintain data consistency and establish relationships between users, construction sites, access requests, and monitoring records.

1. Users and Construction Sites (One-to-Many, 1)

One user can register multiple construction sites, while each construction site is registered by one user. This relationship is established through the created_by field in the construction_sites table, which references users.user_id.

2. Users and Construction Sites through User_Site (Many-to-Many, M)

A user can have access to multiple construction sites, and a construction site can be accessed by multiple users. This many-to-many relationship is managed through the user_site table, which connects users and construction_sites using user_id and construction_id.

3. Users and Site Access Requests (One-to-Many, 1)

A user can submit multiple access requests to view private construction sites. This relationship is established through the request_from_id field in the site_access_request table, which references users.user_id.

A user can also receive multiple access requests from other users. This relationship is established through the request_to_id field, which also references users.user_id.

4. Construction Sites and Site Access Requests (One-to-Many, 1)

A construction site can receive multiple access requests from different users. Each access request is associated with one construction site through the construction_id field in the site_access_request table, which references construction_sites.construction_id.

5. Construction Sites and Progress Logs (One-to-Many, 1)

A construction site can have multiple progress logs, with each log representing a separate monitoring operation. This relationship is established through the construction_id field in the progress_logs table, which references construction_sites.construction_id.

These relationships allow the system to maintain site information, manage user access, process access requests, and store the latest monitoring result for each construction site.

Appendix B — Requirements Traceability Matrix

UFR ID	User Requirement / Summary	Related SFR ID(s)	Acceptance Criteria / Verification	Use Case / Model
FR1	Create a user account.	SFR1	Valid registration details create an account and display a confirmation message.	Sign Up
FR2	Log in using valid credentials.	SFR2	Valid credentials allow login; invalid credentials are rejected.	Login

UFR ID	User Requirement / Summary	Related SFR ID(s)	Acceptance Criteria / Verification	Use Case / Model
FR3	Log out of the account.	SFR3	Logging out ends the session and prevents access to protected pages until login.	Logout
FR4	Store and retrieve user, site, imagery, and monitoring records.	SFR4	Saved records can be retrieved with their stored information intact.	Data Management
FR5	Maintain monitoring operation records and results.	SFR5	A monitoring operation's status and available results are saved and retrievable.	Compare Images; View Monitoring Results
FR6	Associate monitoring records with their corresponding sites.	SFR6	Each monitoring record is linked to the correct construction site.	View Monitoring Results
FR7	Compare satellite images captured at different times.	SFR7	Given suitable images of the same site from different dates, the system compares them and identifies detectable changes.	Compare Images
FR8	Generate a change mask highlighting detected areas.	SFR8	A successful change detection operation produces a mask highlighting detected changes.	Compare Images
FR9	Display original satellite images alongside change detection results.	SFR9	The comparison images and corresponding change mask are displayed with the correct dates.	View Monitoring Results
FR10	Analyze detected changes to provide construction insights.	SFR10	Available change detection results are analyzed and produce relevant findings.	AI Analysis
FR11	Summarize detected changes understandably.	SFR11	The system displays a readable summary based on available monitoring results.	AI Analysis

UFR ID	User Requirement / Summary	Related SFR ID(s)	Acceptance Criteria / Verification	Use Case / Model
FR12	Present AI-generated observations based on image analysis.	SFR12	AI observations are displayed with supporting monitoring data and do not present unsupported conclusions as facts.	AI Analysis
FR13	Display registered sites and monitoring status on a dashboard.	SFR13	The dashboard displays the user's accessible sites and their available monitoring statuses.	View Dashboard
FR14	Display satellite images, detected changes, and measurements.	SFR14	Available images, detected changes, and calculated measurements are displayed for the selected monitoring result.	View Monitoring Results
FR15	Present monitoring results through visual overlays.	SFR15	The system overlays the generated change mask on the corresponding satellite imagery.	View Monitoring Results
FR16	Register a site with its name, coordinates, and construction start date.	SFR16	Valid site details are saved, and the system confirms successful registration.	Register Construction Site
FR17	Validate coordinates and show the location for confirmation.	SFR17	Valid coordinates show the selected location; invalid coordinates are rejected with a message.	Register Construction Site
FR18	View registered construction site details.	SFR18	Selecting a site displays its saved details correctly.	View Construction Site
FR19	Request access using a unique site identifier.	SFR19	A user without access can submit a request for an existing site; duplicate pending requests are rejected.	Request to Join Construction Site
FR20	Allow the site creator to approve or reject access requests.	SFR20	The creator can approve or reject a pending request; only approval grants access.	Manage Site Access Requests

UFR ID	User Requirement / Summary	Related SFR ID(s)	Acceptance Criteria / Verification	Use Case / Model
FR21	View calculated change measurements over time.	SFR21	The system displays available measurements with their corresponding dates and units.	View Monitoring Results
FR22	Compare monitoring results from different dates.	SFR22	Selecting different monitoring dates displays the corresponding results for comparison.	Compare Images; View Monitoring Results
FR23	View measurements from the latest satellite comparison.	SFR23	The latest successfully completed comparison's measurements are displayed for the selected site.	View Monitoring Results
FR24	Generate a report for a selected site and time period.	SFR24	Selecting a site and a valid period with available results produces a report.	Report Generation
FR25	Include images, change detection results, and measurements in reports.	SFR25	The report includes relevant available imagery, change detection results, measurements, and dates.	Report Generation
FR26	Download generated monitoring reports.	SFR26	The user can download and open a successfully generated report.	Report Generation
FR27	Notify users when suitable satellite imagery is unavailable.	SFR27	If no suitable imagery exists for the selected site or period, the system displays a notification.	Compare Images
FR28	Inform users when monitoring cannot proceed because of a processing or service error.	SFR28	When a required service or processing component fails, the system displays an appropriate error message.	Compare Images; Error Handling
FR29	Notify users when site information is invalid or incomplete.	SFR29	Missing required fields or invalid site information prevent registration and produce a validation message.	Register Construction Site

UFR ID	User Requirement / Summary	Related SFR ID(s)	Acceptance Criteria / Verification	Use Case / Model
FR30	Display an error message when monitoring fails.	SFR30	A failed monitoring operation is marked as unsuccessful, the user is informed, and previously completed results remain available.	Compare Images; Error Handling